Introduction to the z-Transform

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Key Concepts

1) The z-transform is a tool for analyzing the interaction of signals and systems.

    a) The z-transform generalizes the discrete-time Fourier transform (DTFT) and is defined for signals that do not have a DTFT.

    b) The z-transform is used to study stability of systems.

    c) The z-transform is used to study causality of systems and the interaction between causality and stability.

d) The z-transform is the discrete-time counterpart to the Laplace transform.

2) The z-transform of a signal $x[n]$ is $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$.

a) z is a complex number, so $X(z)$ is a complex-valued function of a complex-valued variable.

b) The DTFT is obtained from the z-transform by setting $z = e^{j\omega}$.

c) The z-plane has horizontal axis equal to the real part of z and vertical axis equal to the imaginary part of z.

d) The unit circle in the z-plane is $z = e^{j\omega}$ or $|z| = 1$. ω represents the angle of the vector $z = e^{j\omega}$ with respect to the real axis.

e) The z-transform is defined over the z-plane, while the DTFT is defined on the unit circle.

Video Screenshots

Introduction to the z-Transform

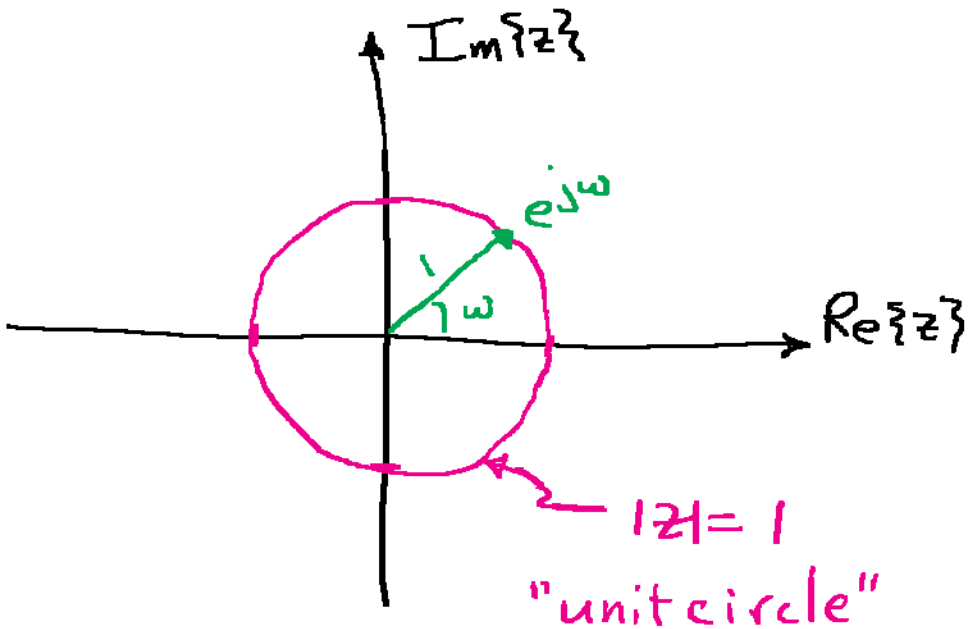
- Generalization of the DTFT
  - signals for which DTFT doesn't exist
  - New ideas: stability causality
- Discrete-time counterpart to the Laplace transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \qquad z \in \mathbb{C} \quad \text{(complex numbers)}$$

DTFT $\quad X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \implies X(e^{j\omega}) = X(z)\big|_{z=e^{j\omega}}$

- Complex valued fcn of real-valued ω
- $X(z)$ is complex-valued fcn of complex-valued z

The Complex Plane (z-plane)



- $X(z)$ is defined on the plane

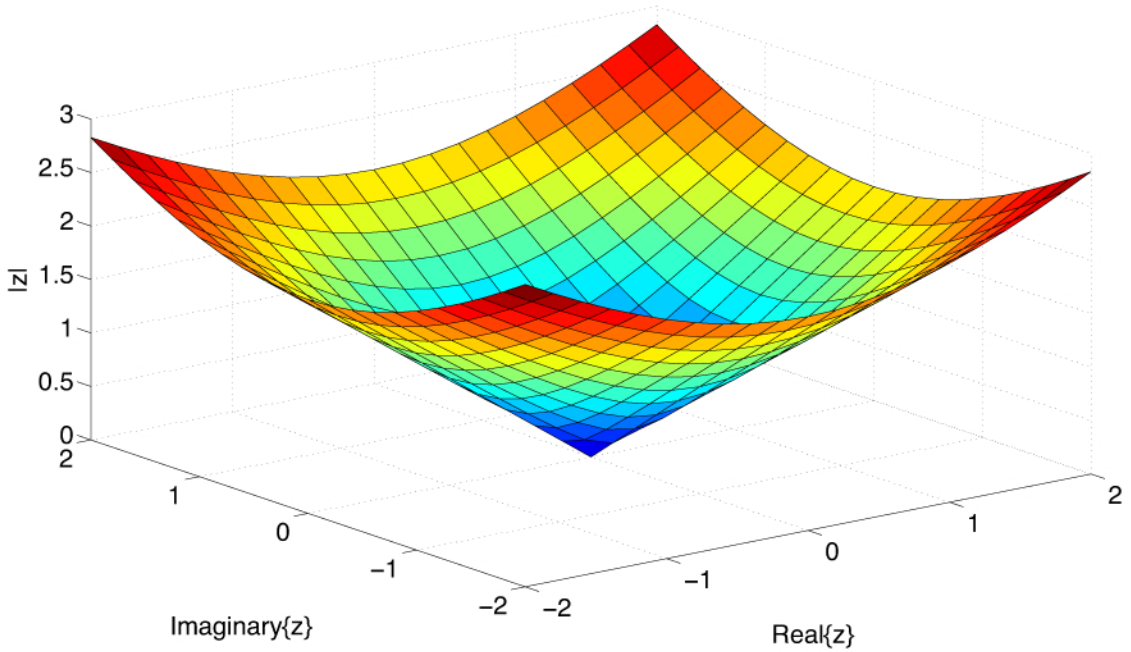
- $X(e^{j\omega})$ is defined only on the unit circle $|z|=1$ or $z=e^{j\omega}$

- As $-\pi < \omega < \pi$, $z = e^{j\omega}$ goes once around the unitcircle
- Confirms 2π periodicity of the DTFT

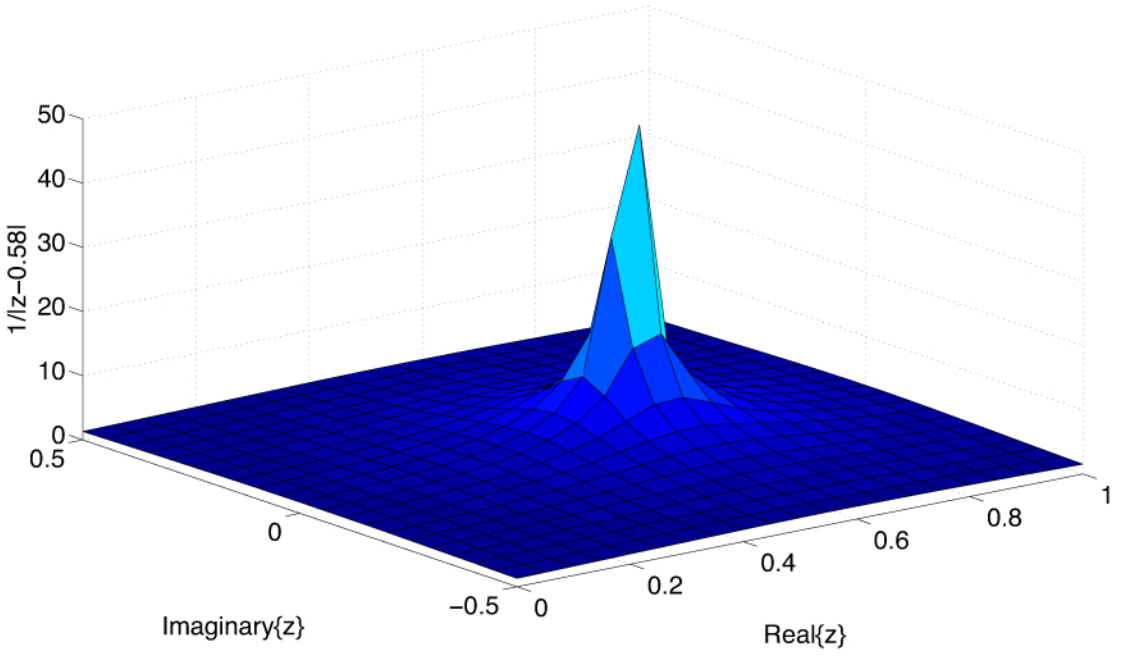
Notation: $x[n] \overset{z}{\longleftrightarrow} X(z)$; $z\{x[n]\} = X(z)$

Examples:

$$X(z) = z$$



$$X(z) = \frac{1}{z - 0.58}$$



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